Air Traffic Control Speaker Diarization with Sortformer: Offline and Streaming Evaluation, Calibration, and Lightweight Domain Adaptation

Abstract—**Air traffic control (ATC) speaker diarization is difficult because Very High Frequency (VHF) radio is band-limited, transmissions are sparse and half-duplex, and a single recording may contain dozens of speakers. This paper evaluates the Sortformer family for ATC diarization on the ATC0 Revised corpus using four held-out recordings from three U.S. facilities. The study jointly examines offline inference, streaming inference, lightweight domain adaptation with a frozen encoder, operating-point calibration, checkpoint sensitivity, and difficulty-stratified behavior under a common segment-level protocol. On the evaluation set, the pretrained offline model reaches a diarization error rate (DER) of 36.6% and a confusion error rate (CER) of 14.4%. A 1000-step fine-tuned offline model reduces these values to 14.9% DER and 6.8% CER. The default streaming configuration lowers CER to 7.7% but remains limited by false alarm (FA), yielding 24.8% DER; selecting a threshold on disjoint calibration recordings and transferring it unchanged to the evaluation set reduces streaming DER to 17.4% and CER to 6.6%. Additional analyses show that checkpoint choice is non-monotonic, low-latency streaming is primarily FA-limited, and fine-tuning is most beneficial in multi-speaker and rapid-turn segments. The main conclusion is that strong ATC diarization with Sortformer is achievable, but only when adaptation, latency, and threshold calibration are treated as coupled design choices rather than isolated implementation details.**

Index Terms—**speaker diarization, air traffic control, Sortformer, streaming inference, domain adaptation, VHF radio**

I. INTRODUCTION

Air traffic control (ATC) communications are safety-critical voice exchanges carried over Very High Frequency (VHF) radio. Automated processing of these communications can support downstream tasks such as transcription, controller–pilot attribution, clearance extraction, traffic-flow analytics, and compliance monitoring [1], [2]. Speaker diarization, namely the task of determining *who spoke when*, is therefore an important upstream component for ATC speech systems.

ATC audio differs substantially from the meeting and telephone speech that dominates diarization benchmarks. The VHF channel limits the effective bandwidth to roughly 300–3400 Hz, removing high-frequency speaker cues [3]. Push-to-talk operation creates sparse, half-duplex exchanges, meaning that local transmissions are effectively one-sided and separated by long silences, and a single frequency can involve far more speakers than conventional diarization corpora. In the ATC0 Revised corpus used here, individual recordings contain 21–90 speakers, well beyond the speaker counts typically encountered in corpora such as CALLHOME and DIHARD [4], [5].

Modern diarization systems have strong results on standard conversational benchmarks, including end-to-end models such as Sortformer [6] and its streaming extension [7], clustering-based pipelines such as pyannote [8], and linear-complexity streaming models such as Long-Form Streaming End-to-End Neural Diarization (LS-EEND) [9]. Their behavior on ATC radio speech is less well characterized. In this domain, acoustic mismatch, extreme speaker cardinality, local speaker-slot ambiguity, and threshold sensitivity can all change the conclusions that would be drawn from a meeting-style evaluation.

This paper studies the Sortformer family as a unified offline and streaming framework for ATC diarization. The goal is not to provide a broad benchmark across every available architecture. Instead, the paper asks a narrower and more practical question: under a clearly specified ATC protocol, how do offline and streaming Sortformer behave before and after lightweight adaptation, and how much of the remaining gap is due to latency or threshold choice rather than the architecture itself? Our main contributions are:

1) We present a controlled evaluation of offline and streaming Sortformer on four held-out ATC recordings from three facilities using 10 s analysis windows, 5 s shift, and 0.25 s collar scoring with overlap retained.
2) We study lightweight domain adaptation through frozen-encoder fine-tuning and show how checkpoint choice changes the DER/CER tradeoff.
3) We quantify the latency–accuracy tradeoff for streaming inference and separate threshold calibration effects from intrinsic model behavior by selecting a common threshold on disjoint calibration recordings and transferring it unchanged to the evaluation set.
4) We complement aggregate DER with difficulty-stratified analysis, speaker-count analysis, role-conditioned CER, per-recording DER decomposition, and cue-level embedding evidence to explain where adaptation helps and where ATC remains difficult.

The remainder of the paper is organized as follows. Section II places the work relative to end-to-end, streaming, and ATC-specific diarization literature. Section III defines the corpus, split construction, model configurations, and scoring procedure. Section IV presents the main quantitative comparisons. Section V examines where the gains come from and which conditions remain difficult. Section VI interprets the results and states the main limitations.

II. Related Work

End-to-end neural diarization (EEND) reformulates diarization as frame-level multi-label prediction with permutation-free objectives [10], [11]. Sortformer [6] replaces permutation-invariant training with a sort-based speaker ordering objective and combines a Neural Error-correcting Self-supervised Transformer (NEST) Fast Conformer encoder [12] with a deep Transformer decoder. Its streaming extension introduces the Arrival-Order Speaker Cache (AOSC), which maintains local speaker state to reduce online ambiguity [7]. LS-EEND uses retention-based sequence modeling to obtain linear temporal complexity for streaming inference [9].

Clustering-based diarization remains important in practice. The pyannote pipeline combines neural speech segmentation, speaker embeddings, and agglomerative clustering [8], [13]. Such systems often benefit from strong embeddings and explicit postprocessing, but their behavior can degrade when bandwidth reduction or radio artifacts distort the acoustic cues that support speaker separation.

Compared with meeting and telephone diarization, the ATC-specific literature is still relatively small and is often coupled to text or role information. Han et al. [3] argued that VHF transmission weakens purely acoustic speaker cues and proposed combining acoustic and textual information. Zuluaga-Gomez et al. [2] showed that speaker-role and speaker-change information can be extracted from ATC transcripts with high accuracy. These works highlight an important point: ATC diarization can benefit from lexical or role context, but such context should not be assumed to be available in all deployments. The present study therefore focuses on acoustic diarization itself and examines how a contemporary offline/streaming diarization model transfers to ATC speech.

III. Experimental Setup

A. Corpus, Annotation, and Data Partitions

We use the ATC0 Revised corpus, a corrected version of the Linguistic Data Consortium (LDC) ATC0 release [14]. In this context, "revised" means that the segment time-marked (STM) transcripts used in this study were corrected for timing, casing, punctuation, and quality annotations before diarization preparation. The corpus contains 16 single-channel VHF recordings from three U.S. facilities and spans approximately 22 hours of audio. Speaker counts range from 21 to 90 per recording.

Ground-truth speaker annotations are derived from the STM files and exported to Rich Transcription Time Marked (RTTM) files. Each STM segment is mapped to an RTTM region with the same start/end time and speaker identifier; no speaker merging or relabeling is applied for diarization scoring. Controllers are identified by position labels containing hyphens (for example, D1-1), while pilots appear as airline or call-sign identifiers. Four recordings are reserved for evaluation: `dca_d1_1`, `dca_d2_2`, `dfw_a1_1`, and `log_id_1`. The remaining 12 recordings are used for fine-tuning. The four evaluation recordings contain 32, 31, 43, and 88 speakers respectively, giving a deliberately varied test set across both site and speaker cardinality.

TABLE I

Data partitions used in this study. Size is the number of windows. The calibration subset is constructed from the first 30 minutes of each non-evaluation recording and is disjoint from the final evaluation set.

Subset	Recordings	Windowing	# Windows
Fine-tuning train	12	90 s windows	1453
Fine-tuning val.	12	90 s windows	456
Calibration	12	10 s, no overlap	2172
Evaluation	4	10 s, 5 s shift	3989

The study uses three disjoint data partitions: a fine-tuning partition, a threshold-calibration partition, and a final evaluation partition. Fine-tuning uses 90 s training windows and contains 1453 training windows plus 456 validation windows. Evaluation uses 10 s analysis windows with 5 s shift, yielding 3989 scored windows in total: 1056 from `dca_d1_1`, 717 from `dca_d2_2`, 781 from `dfw_a1_1`, and 1435 from `log_id_1`. For threshold selection, we create a separate calibration subset from the 12 non-evaluation recordings by taking non-overlapping 10 s windows from the first 30 minutes of each recording. Using non-overlapping windows avoids overweighting highly correlated local segments during threshold selection. This produces 2172 calibration windows (181 per recording) and is used only for threshold selection, never for the final evaluation results.

B. Model Configurations and Fine-Tuning

We evaluate three Sortformer configurations chosen to isolate transfer, latency, and lightweight adaptation.

Pretrained offline Sortformer is the four-speaker offline checkpoint from the NVIDIA NeMo Sortformer release. Inference is performed on 10 s windows with 5 s shift.

Pretrained streaming Sortformer is the corresponding four-speaker streaming checkpoint. In the main comparison we report the model's "medium" latency configuration, corresponding to 10.0 s latency. We also evaluate the "low" and "ultra-low" presets, corresponding to 1.04 s and 0.32 s latency.

Fine-tuned offline Sortformer starts from the pretrained offline checkpoint, freezes the NEST encoder, and updates only the Transformer and output layers for 1000 optimizer steps. In other words, the encoder weights are held fixed while the diarization-specific layers are adapted to ATC speech. This design was chosen to preserve the encoder's broad acoustic priors while allowing the speaker-assignment layers to specialize. Training uses Adam with decoupled weight decay (AdamW), learning rate 10^{-5}, 100-step warmup, batch size 4, and bfloat16 (bf16) precision. Checkpoints are retained every 200 steps for the checkpoint-sensitivity study.

For cue-level embedding analysis, we use TitaNet-Large [15] to compute speaker embeddings from isolated ATC cues, where a cue denotes one contiguous transmission from a single speaker to a listener.

C. Scoring Protocol and Metric Definitions

Sortformer emits up to four *speaker slots*, meaning four output channels that represent locally active speakers within a segment rather than persistent speaker identities across an entire recording. Because those slots are local to each 10 s analysis segment, evaluation is also performed segment by segment rather than after cross-segment identity stitching. Each hypothesis segment is scored against the matching RTTM region using a 0.25 s collar with overlap retained. Here, collar scoring means that boundary errors within 250 ms of a reference boundary are ignored, while overlap retained means that overlapping reference speech is kept in the scoring rather than excluded. Because the 5 s shift creates overlapping evaluation windows, window-level observations are not treated as independent samples; instead, the error components are aggregated by recording. This protocol measures segment-level diarization quality while avoiding an additional speaker-linking stage that is not part of the evaluated system.

Diarization error rate (DER) is decomposed into false alarm (FA), missed speech (MISS), and confusion error rate (CER), so that $\mathrm{DER} = \mathrm{FA} + \mathrm{MISS} + \mathrm{CER}$. FA measures predicted speech where the reference contains no speech, MISS measures reference speech omitted by the system, and CER measures speech assigned to the wrong speaker slot.

For the streaming calibration analysis, we sweep a common onset/offset threshold from 0.40 to 0.90 in steps of 0.05 on the disjoint calibration subset. Threshold calibration here means selecting the posterior threshold that converts streaming probabilities into active-speaker decisions. Threshold selection is based on minimum DER, with FA and then CER used as tie-breakers if needed. The selected threshold is then transferred unchanged to the four evaluation recordings.

The model family supports four output speaker slots. This is not a limitation for the held-out 10 s evaluation windows: none of them contain more than four labeled speakers. However, the 90 s fine-tuning manifests do contain windows with more than four labeled speakers: 13.6% of training windows and 18.9% of validation windows have at least five speakers. This mismatch is important when interpreting adaptation results on long ATC windows.

IV. Results

A. Overall Comparison

Table II summarizes the overall metric decomposition, while Table III reports DER/CER pairs by recording. The pretrained offline model serves as the unadapted baseline and reaches 36.6% DER and 14.4% CER. The default streaming configuration reduces CER to 7.7% but still yields 24.8% DER because FA remains high. The 1000-step fine-tuned offline model gives the strongest overall DER at 14.9% and also reduces CER to 6.8%.

The per-recording view is important because the default streaming gains are not uniform across sites. CER decreases on all four evaluation recordings relative to the pretrained offline model, but on the two DCA recordings FA rises sharply to 29.0% and 32.5%, pushing DER above 43% despite the lower CER. On the DFW and LOG recordings, both DER and CER improve simultaneously. Fine-tuning is the strongest uncalibrated overall system and reduces CER on all four recordings relative to the pretrained offline baseline. Relative CER reductions are 46.3%, 55.9%, 60.7%, and 45.9% on `dca_d1_1`, `dca_d2_2`, `dfw_a1_1`, and `log_id_1` respectively.

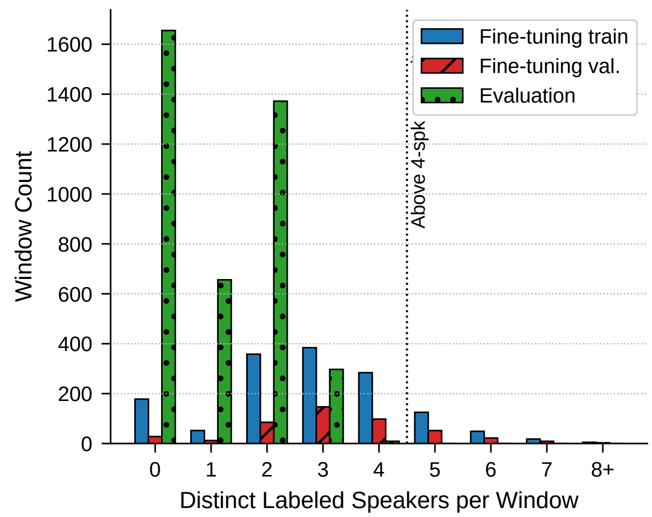

Fig. 1. Speaker-count distribution per window for Fine-tuning train/val. (90 s) and Evaluation (10 s). Evaluation windows never exceed the model's four-speaker output ceiling.

TABLE II
Overall segment-level diarization results on the four evaluation recordings. The fine-tuned offline system gives the best overall DER, while calibrated streaming yields the lowest CER among the streaming configurations discussed in the paper.

System	DER	FA	MISS	CER
Pretrained offline	36.6%	21.2%	1.0%	14.4%
Streaming (10.0 s, default)	24.8%	16.6%	0.5%	7.7%
Streaming (10.0 s, calibrated @ 0.75)	17.4%	5.5%	5.4%	6.6%
Fine-tuned offline (1000 steps)	14.9%	6.3%	1.9%	6.8%

B. Checkpoint Sensitivity

Figure 2 shows performance across the 200, 400, 600, 800, and 1000-step checkpoints from the fine-tuning run. The response is non-monotonic. DER improves sharply from step 200 to step 400, degrades at step 600, and then partially recovers. The best DER occurs at step 400 (14.79%), while the best CER occurs at step 1000 (6.76%), with step 800 nearly tied on both metrics. We report the 1000-step checkpoint as the main adapted offline system because it corresponds to a fixed fine-tuning budget and achieves the lowest CER; if selecting a single checkpoint strictly by DER, step 400 is marginally preferable. The result is practically important: checkpoint selection changes the DER/CER balance and should be reported explicitly rather than treated as a minor training detail.

C. Latency–Accuracy Tradeoff

Figure 3 shows the latency frontier across the three streaming configurations. Reducing latency from 10.0 s to 1.04 s and

TABLE III

Per-recording DER/CER on the evaluation set. Each cell reports DER / CER. The calibrated streaming row uses a threshold selected only on disjoint calibration recordings and transferred unchanged to the evaluation set. The table shows that streaming gains are not uniform across sites and that fine-tuning improves CER on all four recordings.

System	Overall	`dca_d1_1`	`dca_d2_2`	`dfw_a1_1`	`log_id_1`
Pretrained offline	36.6% / 14.4%	20.9% / 16.8%	20.2% / 17.1%	47.4% / 15.9%	44.7% / 10.1%
Streaming (10.0 s, default)	24.8% / 7.7%	43.3% / 13.7%	43.1% / 10.3%	12.4% / 4.2%	15.5% / 6.1%
Streaming (10.0 s, calibrated @ 0.75)	17.4% / 6.6%	25.9% / 12.0%	25.5% / 8.7%	12.1% / 3.6%	13.1% / 5.2%
Fine-tuned offline (1000 steps)	14.9% / 6.8%	12.6% / 9.0%	10.7% / 7.6%	22.1% / 6.3%	11.7% / 5.4%

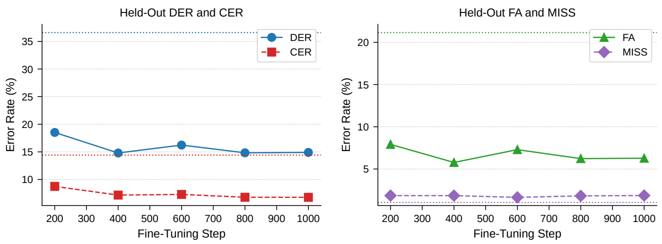

Fig. 2. Checkpoint-response curve for the 1000-step fine-tuning run. The left panel shows DER and CER; the right panel shows FA and MISS. Performance is non-monotonic: the best DER occurs at step 400, whereas the best CER occurs at step 1000.

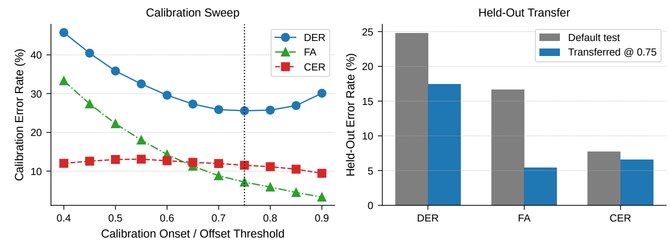

Fig. 4. Streaming threshold sweep on disjoint calibration recordings and unchanged transfer to the evaluation set. The left panel shows the calibration sweep, and the right panel compares default and transferred evaluation metrics. Threshold transfer reduces FA substantially and is therefore central to the interpretation of the streaming results.

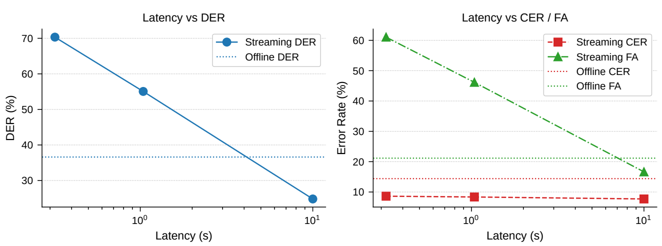

Fig. 3. Streaming latency frontier on the four evaluation recordings. The left panel shows DER versus latency, and the right panel shows CER and FA versus latency. Lower latency increases DER sharply, and the increase is driven primarily by FA rather than by a large increase in CER.

0.32 s increases DER from 24.8% to 55.1% and 70.3%. The main driver is FA, which rises from 16.6% to 46.2% and 61.1%. CER changes much less over the same range, from 7.7% to 8.4% and 8.6%. Thus, the low-latency penalty is dominated by activity overtriggering rather than by a wholesale collapse of speaker assignment.

D. *Threshold Calibration Across Recordings*

The calibration sweep on the 2172-window disjoint calibration subset selects threshold 0.75 by minimum DER. Figure 4 shows both the calibration sweep and the unchanged transfer of that threshold to the evaluation set. Relative to the default streaming configuration, the transferred threshold reduces evaluation DER from 24.8% to 17.4% and FA from 16.6% to 5.5%. CER also improves modestly from 7.7% to 6.6%, while MISS rises from 0.5% to 5.4%. The transferred threshold therefore removes much of the DCA-specific FA inflation without any tuning on the evaluation recordings. After transfer, streaming reaches CER comparable to the fine-tuned offline model (6.6% versus 6.8%), but its overall DER remains higher because MISS is substantially larger.

V. Error Analysis and Supporting Evidence

A. *Difficulty-Stratified Performance*

Figure 5 stratifies the evaluation windows by number of active speakers, number of speaker changes, and speech occupancy. The fine-tuned model roughly halves confusion in the harder multi-speaker bins: speaker-weighted CER falls from 14.96% to 6.95% on two-speaker windows and from 26.42% to 12.55% on three-speaker windows. The same pattern holds for rapid-turn windows: turn-weighted CER falls from 22.21% to 9.94% on windows with three speaker changes. The default streaming model improves CER over the pretrained offline baseline in these bins, but its main weakness is elevated FA during active speech. In the 25–75% speech-occupancy bins, streaming produces roughly 0.95–1.06 s of FA per 10 s window, compared with 0.13–0.16 s for the fine-tuned model. Silent-window hallucination is not the dominant failure mode: default streaming produces less FA on silent windows than the pretrained offline model.

B. *Per-Recording and Auxiliary Analyses*

Figure 6 shows the DER decomposition before and after 1000-step fine-tuning. The improvement is not isolated to one recording: all four evaluation recordings improve in CER, and the DFW and LOG recordings also improve strongly in DER through lower FA. The largest absolute gains combine lower CER with lower FA, especially on the DFW and LOG pair. MISS increases modestly overall, indicating a slightly more conservative operating point after adaptation.

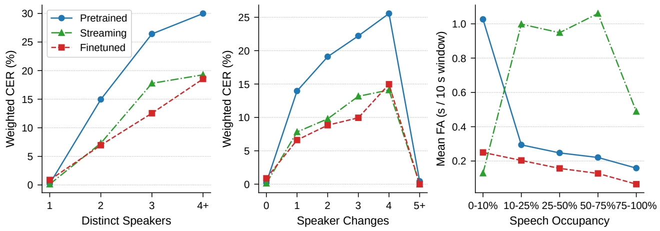

Fig. 5. Difficulty-stratified analysis on the evaluation windows. From left to right, the panels show CER versus distinct speakers, CER versus speaker changes, and mean FA versus speech occupancy. The figure shows that fine-tuning helps most in multi-speaker and rapid-turn segments, while default streaming is limited mainly by FA during active speech.

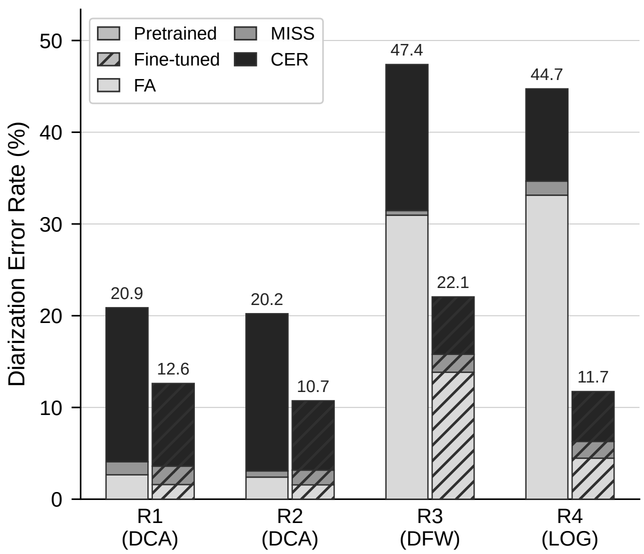

Fig. 6. DER decomposition before and after 1000-step fine-tuning for the four evaluation recordings. The figure helps explain *why* DER improves: fine-tuning reduces CER on all four recordings and lowers FA strongly on the DFW and LOG pair.

Manual inspection of dense controller–pilot exchanges shows a consistent qualitative pattern that agrees with the aggregate results: the pretrained model often keeps one dominant slot active through local turn changes, whereas the fine-tuned model is more likely to activate additional slots around handoff regions. Because any single window-level qualitative example is necessarily illustrative rather than decisive, this observation is treated as secondary to the aggregate evidence in Fig. 5, Fig. 6, and Table III.

Role-conditioned analysis shows that the improvement is not evenly distributed across controllers and pilots. Controller CER falls by 43.4–66.2% across the evaluation recordings, while pilot CER falls by 38.6–49.3%. Controllers occupy 54.5–63.0% of labeled speech time depending on the recording. The larger controller gains are consistent with the hypothesis that controller speech is acoustically more homogeneous within a recording, although the present study does not isolate a causal explanation.

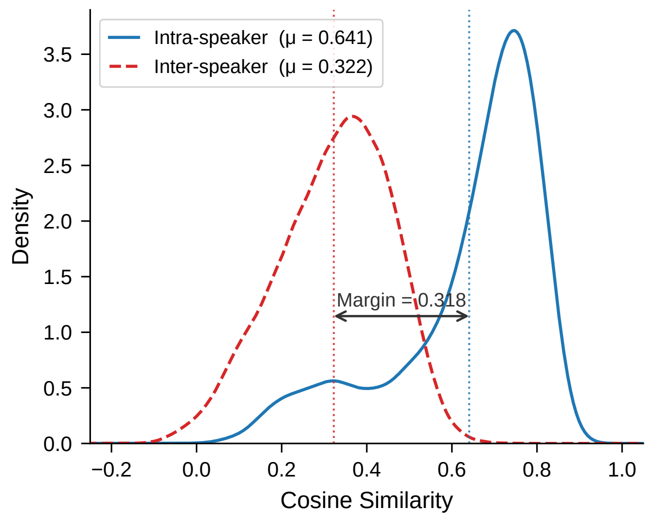

Fig. 7. Cue-level TitaNet-Large similarity distributions for `dca_d1_1`. This auxiliary analysis is not part of the diarization system; it assesses whether VHF ATC audio retains speaker-discriminative cues.

To test whether ATC audio destroys speaker identity information altogether, we extracted TitaNet-Large embeddings from cue-level segments in `dca_d1_1` and `log_id_1`. This embedding analysis is auxiliary evidence and is not used by the Sortformer diarization systems. Figure 7 shows the `dca_d1_1` distribution. The intra-minus-inter cosine similarity margin is 0.318 on `dca_d1_1` and 0.429 on `log_id_1`. Band-limiting the cues to 8 kHz reduces the margin by only 0.02–0.04. These results do not prove that any clustering pipeline will succeed on ATC, but they do indicate that speaker-discriminative information survives the VHF channel to a meaningful extent.

VI. Discussion

The results support four practical conclusions. First, the pretrained offline Sortformer transfers to ATC only partially: missed speech remains low, but confusion and false alarms remain substantial. Second, the streaming architecture improves local speaker assignment, as shown by lower CER on all four evaluation recordings, but its default threshold substantially overfires on the two DCA recordings. Third, frozen-encoder fine-tuning provides an effective and economical adaptation path, yielding the best overall DER while reducing confusion most strongly in multi-speaker and rapid-turn windows. Fourth, both checkpoint choice and streaming latency materially affect the conclusions that would be drawn from the same model family; they should therefore be treated as first-class experimental variables rather than background settings.

Several limitations should also be kept in mind. First, the study evaluates segment-level diarization quality rather than a fully stitched long-form speaker-linking task; a separate cross-window identity reconciliation stage could change the absolute DER values. Second, the evaluation set contains only four recordings, so cross-site generalization should be interpreted cautiously. Third, the model supports four output slots, whereas a non-trivial fraction of the 90 s fine-tuning windows contain more active speakers. Fourth, the calibrated streaming result reflects a single common threshold selected on disjoint calibration recordings; site-specific calibration or adaptive thresholding may yield different operating points in deployment. Operational use in safety-critical avionics contexts would require broader validation and certification-oriented evidence beyond this controlled protocol.

The comparison between the fine-tuned offline and calibrated streaming systems is also instructive. After threshold transfer, the streaming model reaches slightly lower CER than the fine-tuned offline model, indicating that local speaker attribution can be competitive once thresholding is corrected. The offline model nevertheless remains superior in DER because it controls FA and MISS more consistently across sites. In practical terms, applications that emphasize local turn attribution may find calibrated streaming attractive, whereas archive-quality diarization and downstream analytics that penalize insertion and deletion errors more heavily benefit from offline adaptation.

The speaker-count analysis further clarifies why short-window evaluation can be substantially easier than whole-recording speaker counts suggest. Long ATC recordings may involve dozens of speakers, but the local conversational state within a 10 s segment is much simpler. In this setting, the dominant difficulties are rapid speaker changes, short utterances, and threshold stability under band-limited radio audio rather than global speaker cardinality alone. This observation helps explain why a four-slot model remains competitive on 10 s ATC evaluation windows while still being imperfect for longer fine-tuning windows.

The study is intentionally focused on the Sortformer family because it offers matched offline and streaming variants within one architectural framework. Broader comparisons with text-assisted diarization, larger-capacity models, and explicit cross-window speaker linking remain important future directions.

VII. Conclusion

This paper evaluated Sortformer for ATC speaker diarization under matched offline and streaming conditions and showed that both domain adaptation and operating-point selection materially affect performance. Under the common four-recording segment-level protocol, lightweight offline adaptation reduces DER from 36.6% to 14.9% and CER from 14.4% to 6.8%. Streaming improves local speaker attribution, but its operating point is strongly threshold-sensitive: threshold transfer reduces streaming DER from 24.8% to 17.4% and yields CER of 6.6% without any tuning on the evaluation recordings. The checkpoint sweep further shows that adaptation is not monotonic in training steps, while the latency sweep shows that low-latency degradation is driven primarily by false alarms rather than by a large increase in speaker confusion. Overall, the results indicate that strong ATC segment-level diarization with Sortformer is feasible, but reliable deployment requires joint treatment of adaptation, latency, and calibration.

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